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Book Reviews

Generalized Linear Models With Applications in Engineering and the Sciences

Raymond H. Myers, Douglas C. Montgomery, and G. Geoffrey Vining
John Wiley & Sons, New York, NY, 2002, 338 pages, ISBN 0-471-35573-9, $89.95

Non-normal data is commonplace in many engineering related experiments. A traditional approach has been to transform the data and then conduct the analysis using classical normal theory techniques. While data transformation is often very useful, the use of generalized linear models (GLMs) for non-normal data has gained widespread attention in recent years. GLMs make use of the natural distribution of data and accommodate a wide range of non-linear functions when modeling. Many statistical software packages (SAS, SPSS, SPLUS, etc.) have implemented GLMs within their framework. Although the use of GLMs has received a great deal of attention in the biological and biomedical fields for close to 20 years, only recently have engineering disciplines begun to embrace these techniques (Hamada and Nelder, 1997; Myers and Montgomery, 1997).

This is the first book on GLMs written by authors who are primarily associated with engineering related applications. Combined, the authors have over 85 years of experience in statistical consulting and training. All three authors have produced well-received texts in such areas as experimental design, modeling, and engineering statistics. Throughout the book, the authors continually keep in mind that their readers are students of statistics, whether this may be practitioners, graduate students, or even Ph.D. level statisticians who desire familiarity with an area outside of their specific expertise.

The text begins with a nearly 100-page review of linear and non-linear regression. Although it is stated in the preface that the material is appropriate for anyone who has completed a course in regression analysis, the essentials of regression analysis are thoroughly covered in the first three chapters. Chapter 2, "Linear Regression Models", is 50 pages and covers material on estimation (least squares, weighted least squares, and maximum likelihood), inference, and diagnostic checking. The topical format of Chapter 2 "Estimation, Inference, and Diagnostic Checking" could be viewed as a template for Chapters 3–6. The benefit of this is that the reader is afforded the opportunity of understanding the intuitive genesis of GLMs. As with all of the topics covered in the text, the authors illustrate concepts with well-chosen examples.

When encountered with a non-linear model, a natural approach may be to simply transform the model into a linear model and analyze the transformed model. Recognizing this, the authors begin Chapter 3, "Nonlinear Regression Models," by taking the time to show the reader the potential perils of doing this. It would be easy enough for the authors to explain this as multiplicative versus additive errors and then "leave it to the reader" to figure out the subtleties. However, the authors take a data set and analyze it both ways, showing the subtleties. Another highlight of Chapter 3 arrives in the explanation of parameter estimation in non-linear regression. It is well known that for non-linear regression, when the parameters are estimated via least squares, the score equations have no simple closed-form solutions and some iterative technique is used to find a solution. This is clear when one works out the messy derivatives involved with least squares of a non-linear model, but often, intuition stops here. The authors once again appeal for a higher intuition and give a nice explanation of the geometry of linear versus non-linear least squares. For illustrative purposes, they consider a two variable linear and non-linear model and then show the sum of squared residuals contour plot versus the two model parameters. It is clear why a unique solution exists for least squares of linear model but not why non-unique solutions exist in the non-linear model setting. The illustration is clear and sets the stage for complicated design topics discussed in the final chapter.

Logistic and Poisson regression are perhaps the most common applications of GLMs. Chapter 4, "Logistic and Poisson Regression Models", is devoted to the discussion of these models. In the section on estimation, the authors not only relate the discussion back to estimation in the classic linear model but they also demonstrate that maximum likelihood and weighted least squares result in the same score equation, thereby providing the same parameter estimates. Regarding inference, a careful explanation is provided on the differences between Wald and likelihood inference. Example data sets are analyzed throughout the chapter using SAS PROC LOGISTIC and PROC GENMOD. One particularly interesting data set deals with an electronic circuit card assembly by a wave-soldering process. The response is Poisson and is characterized by overdispersion. Ignoring overdispersion can result in overly optimistic standard errors of the parameter estimates. This concept is well-illustrated as the authors analyze the data

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two ways: (i) naively, by ignoring the overdispersion; and (ii) accounting for the overdispersion. As with several examples in the book, it would be helpful if the authors had provided a section containing the actual code used for analyses.

In Chapter 5, "The Family of Generalized Linear Models," the formal definition of a GLM is presented. Likelihood expressions are now developed in a more general sense but careful selection of notation enables the reader to draw parallels to material discussed in previous chapters. A critical component of successfully using GLMs is the ability of the user to correctly specify the link function. Strategies for choosing the appropriate link function are rarely discussed in other GLM texts. The authors provide an interesting section (Section 5.6) on the use of power functions to determine the link. The chapter ends with two excellent examples on the use of the gamma distribution.

There are many situations in which data is collected in a repeated measures format. For example, in longitudinal studies, experimental units are measured repeatedly over time. Other examples include experimentation in which some variables are difficult/costly to change. Whenever repeated measures are taken, the responses from one measure to the next are inherently correlated. This presents problems for the GLM framework, which assumes independent responses. The authors take issue with these types of experiments in Chapter 6, "Generalized Estimating Equations". After the methodology is introduced and discussed, several examples follow illustrating their use. The authors point out that the robust standard error estimates are problematic with small sample sizes and an intuitive adjustment for standard errors is explained. Again SAS PROC GENMOD is used for data analysis and the examples are well chosen as they illustrate the perils of ignoring the existence of correlation. Generalized linear mixed models are not discussed.

The final chapter of the text, "Further Advances and Applications in GLM" is somewhat theoretical and gives the reader a sense of wanting more information. Experimental designs for GLMs, asymptotic results, the analysis of screening experiments, data transformation, modeling for both the process mean and variance, and generalized additive models are the topics discussed. While much of the material in this chapter could be thought of as "on the cutting edge" of research, there has been some interesting work done and with a little effort, material in this chapter could be expanded in future editions of the book.

I recommend this text to anyone who has an interest in the analysis of non-normal data. The book is an excellent reference for both the practitioner and the academic alike. There are very few authors who can take a relatively complex topic and keep the material from intimidating the readers. The organization, notation, and illustrative examples are well thought out, making this a fun text to read.

References

Hamada, M. and Nelder, J.A. (1997) Generalized linear models for quality-improvement experiments. *Journal of Quality Technology*, **29**, 292–304.

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Contributed by the Book and Software Review Department

Building and Solving Mathematical Programming Models in Engineering and Science

E. Castillo, A.J. Conejo, P. Pedrigal, R. Garcia, and N. Alguacil

John Wiley & Sons, 2002, 546 + 16 pages, ISBN 0-471-15043-6

This book is intended for engineers and applied scientists who need to solve real-world problems using mathematical programming models. The exercises at the end of each chapter also make the book suitable as a textbook in graduate-level courses. Familiarity with matrix algebra, calculus, and analytical geometry (e.g., cones and convex sets) is essential for a full understanding of this book. This book emphasizes the understanding of the issues involved in formulating and solving mathematical models rather than technical algorithmic details. Numerous polished graphical illustrations and practical examples are used to achieve this objective. The book is generally well written. It would enhance the book if the authors create a web site for the book in which they include such things as errata and solutions to selected exercises.

The book contains 13 chapters and two appendices. The chapters are organized in four parts: (I) models; (II) methods; (III) software; and (IV) applications. Part I discusses how mathematical programming models are formulated from real-world problems. These mathematical programming models include:

- Chapter 1: linear programming is introduced by several illustrative examples.
- Chapter 2: mixed-integer linear programming, including binary variables, is also introduced by illustrative examples.